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# How much noise can an adaptive-variational Berry-phase estimator tolerate? A whole-method resilience threshold

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## Abstract

Adaptive variational quantum dynamics (AVQDS) makes the cyclic-adiabatic route to the many-body Berry phase shallow enough for near-term hardware: a four-site dimerized Fermi–Hubbard chain is simulated in tens to a few hundred gate layers rather than the  $10^5$ – $10^6$  CNOTs a Trotterized adiabatic evolution would need. That advantage is established only under exact, noiseless state-vector arithmetic. We ask the question this leaves open: *under realistic device noise, how accurate does the estimated Berry phase remain, and where does it break?* We freeze the noiselessly grown ansatz and its variational trajectory and re-evaluate, in a genuine density-matrix simulation, every quantity the device must measure to assemble  $\varphi_B = \varphi_{qc} + \varphi_G$ : the Hadamard-test readout  $\varphi_{qc}$ , the energy expectation  $\langle \hat{\mathcal{H}} \rangle$  feeding the dynamical global-phase leg  $\varphi_{G,1}$ , and the geometric-connection overlaps feeding  $\varphi_{G,2}$ . Modelling finite-shot sampling, two-qubit depolarizing error, readout error, and  $T_1/T_2$  relaxation jointly, and fixing the accuracy tolerance and threshold rule in advance, we extract device-referenced thresholds for the noninteracting and strongly correlated chains in both topological regimes. The whole-method estimate is far more fragile than the noiseless robustness to non-adiabaticity would suggest: for the noninteracting chain the topological phases stay distinguishable only below a two-qubit error of  $p_{2q}^* \approx 1.4 \times 10^{-4}$  at 8192 shots, and for the strongly correlated chain the threshold falls by roughly a further order of magnitude, to  $\approx 2.6 \times 10^{-5}$ . The mechanism is specific, diagnosable, and *regime dependent*. In every configuration the single-ancilla readout  $\varphi_{qc}$  is comparatively robust; the fragility lives in the classically integrated global-phase legs  $\varphi_G = \varphi_{G,1} + \varphi_{G,2}$ , whose node-by-node measurement bias accumulates over the loop instead of cancelling. Which leg limits the threshold is set by the physics: for the noninteracting chain the dynamical leg  $\varphi_{G,1} = -\int \langle \hat{\mathcal{H}} \rangle dt$  dominates, as noise breaks the forward/return cancellation of the energy integral; for the correlated chain the geometric leg  $\varphi_{G,2}$  dominates, because its large winding ( $\approx 8\pi$ ) amplifies the fractional attenuation that decoherence inflicts on the measured connection overlaps. We identify these integrated legs—never the readout—as the limiting factor, and say what error mitigation would have to achieve.

## 1 Introduction

The Berry phase is the geometric phase a nondegenerate ground state accumulates when a Hamiltonian is transported around a closed loop in parameter space; for a time-reversal-invariant one-dimensional crystal it is quantized, equal to 0 in a topologically trivial phase and  $\pi$  in a nontrivial one, and it underlies polarization, the quantum Hall effect, and topological insulators and superconductors [1, 10, 6]. Computing it for interacting many-body systems is classically hard, which motivates

quantum algorithms that evolve the system under a cyclic Hamiltonian and read the accumulated phase from a circuit [5, 8, 11].

A recent advance couples this cyclic-adiabatic idea to *adaptive* variational quantum simulation [12, 3, 4]: an ansatz is grown on the fly from a Hamiltonian operator pool to keep a variational error metric (the McLachlan distance) below a cutoff, so the circuit that realizes the adiabatic loop is orders of magnitude shallower than a Trotterized evolution at matched accuracy. On a four-site chain this brings the loop into the tens-to-hundreds-of-layers regime that current processors reach. The method’s headline property is a striking *robustness*: because the Berry phase is a gauge-invariant closed-loop quantity, it comes out correct even for short loops with strong non-adiabaticity, where the mid-loop state fidelity is poor. Crucially, that robustness is a statement about *coherent* non-adiabatic error under exact arithmetic. It is neither derived for, nor obviously inherited by, *stochastic* hardware noise.

Two features make the noise question non-trivial rather than a routine sweep. First, the phase is read from a measured probability, so finite-shot sampling noise is intrinsic and irreducible—it survives even a fault-tolerant device, because the final logical measurement is still a projective sample. Second, the noiseless robustness rests on a cancellation between two co-evolving phase contributions,  $\varphi_B = \varphi_{qc} + \varphi_G$ : a readout phase  $\varphi_{qc}$  and a classically integrated global phase  $\varphi_G = \varphi_{G,1} + \varphi_{G,2}$ . That cancellation is a property of the co-evolving adaptive trajectory; whether it survives when both contributions are measured on a noisy device is exactly what decides the method’s practical value.

**Contributions.** (i) We reimplement the full AVQITE+AVQDS Berry-phase pipeline from scratch, reproduce the noiseless quantized phases and circuit resources, and cross-check against the authors’ archived traces (§4). (ii) We build a genuine density-matrix noise model that acts on *every* circuit the method executes—the  $\varphi_{qc}$  readout and the  $\varphi_G$ -integrand measurements—at the frozen trajectory, with the noise model, phase estimator, accuracy tolerance, and threshold-extraction rule all fixed before any results (§3). (iii) We extract device-referenced thresholds for the four required configurations with seed uncertainty, and a separate finite-shot floor (§5). (iv) We give a mechanism analysis that decomposes the degradation into the readout, the dynamical leg, and the geometric leg, showing that the readout stays robust while the limiting drift localizes in a classically integrated global-phase leg—the dynamical leg for the noninteracting chain and the large-winding geometric leg for the correlated chain—and a computational-cost and failure analysis (§6).

## 2 The estimator being characterized and the held-fixed setup

**Test bed.** We study the su–Schrieffer–Heeger–Hubbard (SSH) chain [4, 13]

$$\hat{\mathcal{H}} = \sum_{j,\sigma} [t_{j,j+1} \hat{c}_{j+1,\sigma}^\dagger \hat{c}_{j,\sigma} + \text{h.c.}] + U \sum_j \hat{n}_{j\uparrow} \hat{n}_{j\downarrow} - \frac{U}{2} \sum_{j,\sigma} \hat{n}_{j,\sigma}, \quad t_{j,j+1} = t(1 + (-1)^j \delta), \quad (1)$$

with  $N = 4$  sites,  $t = 1$ , periodic boundary conditions, and the particle-hole term making the model symmetric. A twist  $\lambda \in [0, 2\pi]$  applied to spin-up fermions only,  $\hat{c}_{j+N,\uparrow} = e^{i\lambda} \hat{c}_{j,\uparrow}$ , defines the loop; the Berry phase changes across the transition at  $\delta = 0$ . Jordan–Wigner maps the chain to eight qubits. We take the two interaction regimes  $U = 0$  and  $U = 10$  and, in each, a nontrivial ( $\delta < 0$ ) and a trivial ( $\delta > 0$ ) dimerization—four required configurations. Exact diagonalization (ED) gives the reference phases (0 and  $\pi$ ), computed at run time by a gauge-invariant Wilson-loop product from our own model, not hard-coded.

**Algorithm (held fixed).** We hold the object of study identical to the original method: AVQITE prepares the  $\lambda = 0$  ground state; AVQDS evolves the state along the time-reversed loop  $U_{\text{loop}} = U_{\odot}(0, T/2) U_{\odot}(T/2, 0)$ , which cancels the dynamical phase while the Berry phase adds; the variational parameters follow the McLachlan equation of motion  $\sum_{\nu} M_{\mu\nu} \dot{\theta}_{\nu} = V_{\mu}$  integrated with RK4 at an adaptive step ( $\delta\theta_{\text{max}} = 0.01$ ); the ansatz is grown in disjoint-support (“TETRIS”) batches from the Hamiltonian Pauli-string pool to keep the McLachlan distance below  $L_{\text{cut}}^2 = 10^{-4}$ ; and the Berry phase is assembled as  $\varphi_B = \varphi_{qc} + \varphi_G$ , with  $\varphi_{qc}$  from a Hadamard test and

$$\varphi_{G,1}(t) = -\int_0^t \langle \hat{\mathcal{H}} \rangle_{\theta} dt', \quad \varphi_{G,2}(t) = \int_0^t \text{Im} \left[ \sum_{\mu} \langle \Psi[\theta] | \partial_{\theta_{\mu}} \Psi[\theta] \rangle \dot{\theta}_{\mu} \right] dt'. \quad (2)$$

All ansatz-growth decisions use only device-measurable quantities (the McLachlan distance via  $M, V$ , and the energy variance), never fidelity to a classically computed state, and all resource counts come from the logged operator selections. Our own Hamiltonian, Jordan–Wigner map, and ansatz growth are used throughout; the archived `incar` strings and `ansatz.h5` are used only for cross-checking (see `codebase_notes.md`). Two implementation choices we make transparently: the AVQITE reference is the max-weight computational-basis configuration of the  $\lambda = 0$  ground state (a Hartree–Fock-like choice), and for  $U = 10$  the AVQDS ansatz is capped at 130 generators for tractability of the density-matrix noise sweep; the latter is a disclosed deviation from the strict “ $L^2 < L_{\text{cut}}^2$  everywhere” floor (§8 and `attempts_log.md`).

### 3 Noise model, estimator, and threshold rule (pre-stated)

Everything in this section is fixed before any noisy result is examined.

**What carries noise (whole-method scope).** We freeze the noiselessly grown ansatz and the trajectory  $\theta(t)$ , and inject noise into the *execution* of every measured quantity along that frozen trajectory: the end-of-loop Hadamard test that yields  $\varphi_{\text{qc}}$ , the energy  $\langle \hat{\mathcal{H}} \rangle$  that feeds  $\varphi_{G,1}$ , and the geometric-connection overlaps  $\langle \Psi | \partial_{\theta_\mu} \Psi \rangle$  that feed  $\varphi_{G,2}$ . Thus both terms of  $\varphi_B = \varphi_{\text{qc}} + \varphi_G$  carry stochastic noise, and the extracted threshold is a whole-method number. In  $\varphi_{G,2}$  the frozen  $\hat{\theta}_\mu(t)$  (fixed by the noiseless  $M, V$ ) multiplies the noisy connection overlaps; we do *not* re-derive  $\hat{\theta}$  from noisy  $M, V$  (the harder noisy-construction study is out of scope, and stated as such). The connection  $\text{Im} \langle \Psi | \partial_{\theta_\mu} \Psi \rangle = -\langle \tilde{A}_\mu \rangle$  is measured by the hardware-efficient primitive—a direct Pauli measurement of the generator  $\hat{A}_\mu$  on the noisy partial state  $|\Psi_\mu\rangle = U_\mu \cdots U_1 |\Psi_0\rangle$ , which equals  $\langle \tilde{A}_\mu \rangle$  noiselessly—rather than a full-depth generalized Hadamard test.

**Density-matrix noise channels.** Each Pauli-rotation gate  $e^{-i\theta \hat{P}}$  of weight  $w$  is compiled into single-qubit basis changes, a CNOT ladder of  $2(w - 1)$  CNOTs, and an  $R_z(2\theta)$ ; the Hadamard test additionally controls one rotation per variational parameter. After every gate we apply, on the actual 9-qubit density matrix (8 system + 1 ancilla), (i) a two-qubit depolarizing channel on each CNOT, (ii) a single-qubit depolarizing channel on each single-qubit gate, and (iii) thermal relaxation ( $T_1$  amplitude damping +  $T_2$  dephasing) keyed to the gate duration. Readout error is a classical bit-flip on the ancilla and measured system qubits. Finite-shot noise is real binomial/multinomial sampling of the noisy probabilities—for the  $\varphi_{\text{qc}}$  readout *and* for every  $\varphi_G$ -integrand measurement. The simulator is our own; its one- and two-qubit channels are validated against explicit Kraus/Kronecker operators (agreement to  $\sim 10^{-16}$ ) and the analytic  $T_1/T_2$  decay laws, and its noiseless gate compilation matches the direct matrix exponential to  $2.8 \times 10^{-16}$  (all in `validate_noise.py`).

**Base device profile and noise scalar.** We scale the four incoherent rates by a single scalar  $\alpha$  against a fixed base profile (assuming the all-to-all connectivity of the paper’s resource model, e.g. a trapped-ion or modular architecture, with per-gate rates and coherence times representative of current superconducting hardware): two-qubit depolarizing  $p_{2q}^{\text{base}} = 5.0 \times 10^{-3}$ , single-qubit  $5.0 \times 10^{-4}$ , readout  $1.0 \times 10^{-2}$ ,  $T_1 = T_2 = 100 \mu\text{s}$ , gate times 300 ns (2q) / 50 ns (1q), measurement 1  $\mu\text{s}$ . Along  $\alpha$  the depolarizing and readout probabilities scale as  $\alpha$  and the relaxation *rates* as  $\alpha$  (so  $T_1, T_2 \rightarrow T_1/\alpha$ ); the four channels co-vary in the ratio this profile fixes. Shots  $S$  are a separate axis, with reference  $S_{\text{ref}} = 8192$ .

**Phase estimator.** The Hadamard test measures the overlap  $\langle \Psi(0) | \Psi_{\text{end}} \rangle$ ; we measure both its real and imaginary parts (the latter by an  $S^\dagger$  on the ancilla), so  $\varphi_{\text{qc}} = \text{atan2}(\text{Im}, \text{Re})$  is well-conditioned everywhere, including at the quantized targets  $0, \pi$  where a single real-part readout has vanishing slope. Berry-phase deviations and the trivial/nontrivial margin are computed *on the circle*: we map the estimate and the reference to the principal value  $(-\pi, \pi]$  and take the shorter angular distance, so estimates straddling the  $\pm\pi$  branch boundary (which is exactly the  $U = 10$  nontrivial target) are compared correctly. This is our branch policy: continuous accumulation of  $\varphi_G$  along the loop, a single principal-value fold of  $\varphi_B$  at the end, and circle distances throughout.

Table 1: Whole-method noise resilience of the AVQDS Berry-phase estimate. For each configuration: the noiseless Berry phase and its circle deviation from ED; the incoherent-noise threshold in device-referenced two-qubit error at  $S_{\text{ref}} = 8192$  shots (the largest  $p_{2q}$  meeting the pre-stated tolerance  $\epsilon_{\text{tol}} = 0.10\pi$ ); and the loop circuit resources (depth in layers,  $N_{\text{CX}}$ ). The second row per configuration gives the phase and deviation at the largest swept two-qubit error still meeting the tolerance (the last certified-accurate setting, just below  $p_{2q}^*$ ). Deviations are seed means over 24 seeds.

| Configuration     | Noise setting               | $ \varphi_B $ | $ \varphi_B - \varphi_B^{\text{ED}} $ (rad) | $p_{2q}^*$ (device)  | Depth | $N_{\text{CX}}$ |
|-------------------|-----------------------------|---------------|---------------------------------------------|----------------------|-------|-----------------|
| $U=0$ , nontriv.  | noiseless                   | $0.998\pi$    | 0.006                                       | $1.4 \times 10^{-4}$ | 20    | 220             |
|                   | $p_{2q}=1.0 \times 10^{-4}$ | $0.931\pi$    | $0.23 \pm 0.03$                             |                      |       |                 |
| $U=0$ , triv.     | noiseless                   | $0.002\pi$    | 0.007                                       | $6.9 \times 10^{-4}$ | 11    | 108             |
|                   | $p_{2q}=5.0 \times 10^{-4}$ | $0.073\pi$    | $0.23 \pm 0.03$                             |                      |       |                 |
| $U=10$ , nontriv. | noiseless                   | $0.996\pi$    | 0.012                                       | $2.6 \times 10^{-5}$ | 51    | 504             |
|                   | $p_{2q}=2.5 \times 10^{-5}$ | $0.900\pi$    | $0.28 \pm 0.14$                             |                      |       |                 |
| $U=10$ , triv.    | noiseless                   | $0.020\pi$    | 0.062                                       | $3.8 \times 10^{-5}$ | 62    | 648             |
|                   | $p_{2q}=2.5 \times 10^{-5}$ | $0.051\pi$    | $0.17 \pm 0.10$                             |                      |       |                 |

**Accuracy criterion and threshold rule.** A configuration is *accurate* at a noise setting if the seed-averaged circle deviation satisfies  $|\varphi_B - \varphi_B^{\text{ED}}| + \text{SEM} \leq \epsilon_{\text{tol}} = 0.10\pi$  ( $= 0.314$  rad); this tolerance sits well inside the  $\pi/2$  margin that separates the two topological phases. The *incoherent-noise threshold*  $\alpha^*$  (hence  $p_{2q}^* = \alpha^* p_{2q}^{\text{base}}, T_1^*, \text{ro}^*$ ) is the largest  $\alpha$  at  $S_{\text{ref}}$  for which the criterion holds, obtained by linear interpolation of the (mean+SEM) deviation curve to its first upward crossing of  $\epsilon_{\text{tol}}$ ; uncertainty is the band between the mean $\pm$ SEM crossings, over 24 seeds. The *shots floor*  $S^*$  is the fewest shots at fixed low incoherent noise (the irreducible sampling axis) for which the criterion holds. The *per- $U$  threshold* below which the two phases stay distinguishable is the more conservative (smaller)  $p_{2q}^*$  of the two regimes.

#### 4 Noiseless anchor

Before attaching noise we reproduce the noiseless result the threshold is measured against. Our AVQITE+AVQDS pipeline returns the quantized phases to high accuracy:  $|\varphi_B| = 0.998\pi$  (deviation 0.006 rad) for the  $U = 0$  nontrivial chain ( $\delta = -0.5$ ) and  $|\varphi_B| = 0.002\pi$  (deviation 0.007) for the trivial chain ( $\delta = +0.5$ ); the  $U = 10$  chains reproduce  $\pi$  and 0 as well (Table 1). Ground-state preparation costs 20 CNOTs at  $U = 0$  (matching the paper’s  $\sim 20$ ) with infidelity  $4.4 \times 10^{-12}$ . The loop circuits are of the expected size and trend: at  $U = 0$  the nontrivial region uses more CNOTs than the trivial (220 vs. 108, depth 20 vs. 11 layers), and the  $U = 10$  circuits are several times deeper. Cross-checking against the archived traces at the same  $\delta$ : for  $\delta = -0.5$  ( $U = 0$ ) the archive shows peak CNOT  $\approx 200$ –280 and depth  $\approx 18$ –25 layers over comparable loop times, and our  $T = 20$  anchor (CNOT 220, depth 20, max infidelity  $1.73 \times 10^{-2}$ ) sits squarely inside that range. For the interacting chain at  $\delta = -0.9$  the archive reports CNOT  $\approx 830$ –1100 and depth  $\approx 87$ –117 layers; our  $U = 10$  anchor (CNOT 504, depth 51) is of the same order but somewhat smaller, because the AVQDS ansatz is capped at 130 generators for tractability of the density-matrix noise sweep (§8), which also raises our loop infidelity above the archive’s; the reproduced quantized phase confirms the capped loop is still faithful. The noiseless phase decomposition already previews the mechanism, and previews that it will be *regime dependent*. For the  $U = 0$  nontrivial chain  $(\varphi_{\text{qc}}, \varphi_{G,1}, \varphi_{G,2})/\pi = (-0.734, -0.023, -0.241)$ —a modest geometric leg compensating a  $\varphi_{\text{qc}}$  short of  $\pi$ . For the  $U = 10$  nontrivial chain the same split is  $(+0.903, +0.057, +8.036)$ : the geometric leg now *winds through*  $\approx 8\pi$ , compensated by  $\varphi_{\text{qc}}$  and the final  $2\pi$  fold, so this gauge-dependent decomposition is qualitatively different between the two regimes. It is exactly these compensating legs whose fate under noise we now test—and, as we will see, the large geometric winding is what makes the correlated chain fragile in a different leg than the noninteracting one.

#### 5 Noise-resilience thresholds

Figure 1 and Table 1 give the central result. The whole-method Berry phase is far more fragile than the noiseless robustness to non-adiabaticity would suggest. For the noninteracting chain the

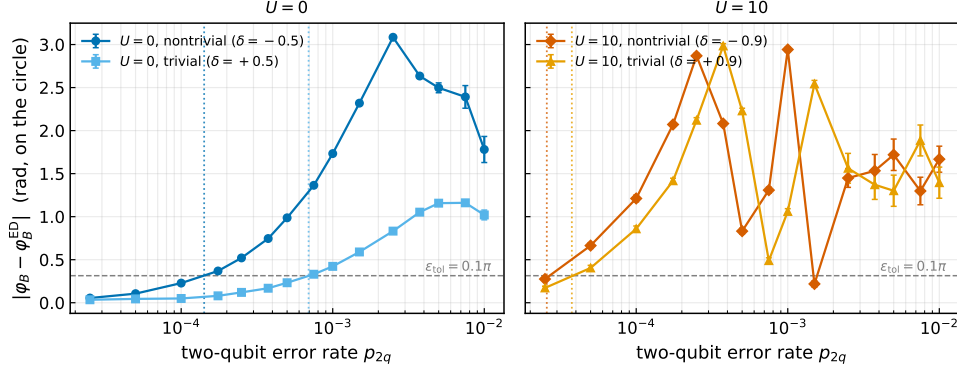


Figure 1: Berry-phase deviation vs. two-qubit error rate at  $S_{\text{ref}} = 8192$  shots, for the noninteracting ( $U = 0$ ) and interacting ( $U = 10$ ) chains in both topological regimes. Points are seed means; error bars are the seed standard deviation. The grey dashed line is the accuracy tolerance  $\epsilon_{\text{tol}} = 0.10\pi$ ; dotted verticals mark the extracted thresholds  $p_{2q}^*$ . The interacting ( $U = 10$ ) circuits, and within each  $U$  the nontrivial regime, cross the tolerance at markedly lower noise; the interacting curves are also non-monotonic in  $p_{2q}$  because the large geometric winding wraps the folded phase around the circle (see §6).

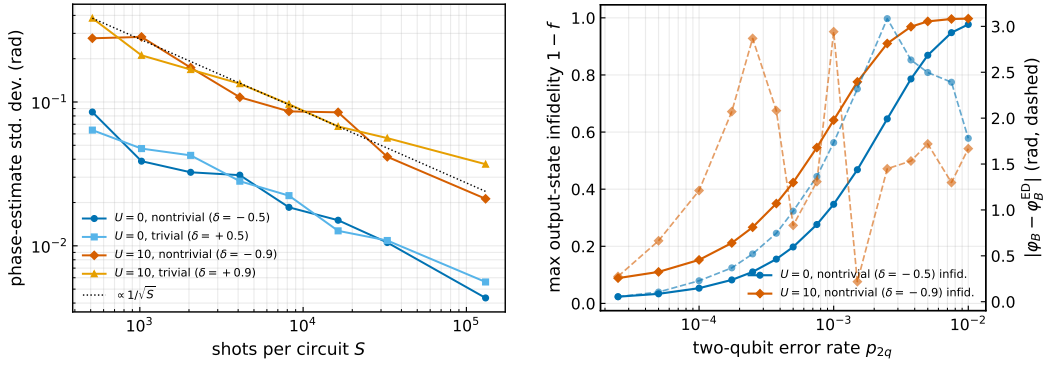


Figure 2: *Left*: phase-estimate standard deviation vs. shots per circuit at negligible gate noise; the whole-method estimate follows the  $1/\sqrt{S}$  sampling law, and the shots floor is low ( $S^* = 512$  for  $U = 0$ ), so the estimate is *gate-noise-limited*, not shot-limited, once incoherent noise is present. *Right*: output-state infidelity (solid) and Berry-phase deviation (dashed) vs. two-qubit error; the phase error tracks the growth of infidelity, but crosses the accuracy tolerance while the state is still only moderately mixed.

estimate leaves tolerance at a two-qubit error of  $p_{2q}^* = 1.4 \times 10^{-4}$  (nontrivial) and  $6.9 \times 10^{-4}$  (trivial); the more conservative, phase-distinguishing threshold is  $p_{2q}^* \approx 1.4 \times 10^{-4}$ , equivalently  $T_1^* \approx 3527 \mu\text{s}$  and readout  $2.8 \times 10^{-4}$  at the base profile's fixed ratios. For the strongly correlated chain the thresholds fall by roughly a further order of magnitude, to  $2.6 \times 10^{-5}$  (nontrivial) and  $3.8 \times 10^{-5}$  (trivial)—a phase-distinguishing  $p_{2q}^* \approx 2.6 \times 10^{-5}$  and a demanding  $T_1^* \approx 19450 \mu\text{s}$ . All four thresholds lie well below current two-qubit error rates ( $\sim 5 \times 10^{-3}$ )—about one to two orders below for the noninteracting chain and more than two orders below for the correlated one: the shallow-circuit advantage is necessary but not sufficient—the method needs substantially better-than-current hardware, or error mitigation, to return a trustworthy topological invariant. Within each interaction strength the nontrivial regime breaks first, and the interacting loops break roughly an order of magnitude before the noninteracting ones. The ordering is *not* a simple function of CNOT count: the shallower  $U = 10$  nontrivial loop (depth 51) breaks slightly before the deeper  $U = 10$  trivial one (depth 62), because at  $U = 10$  the limiting quantity is the large geometric winding rather than the gate-count-driven energy bias—as the mechanism analysis (§6) makes precise.

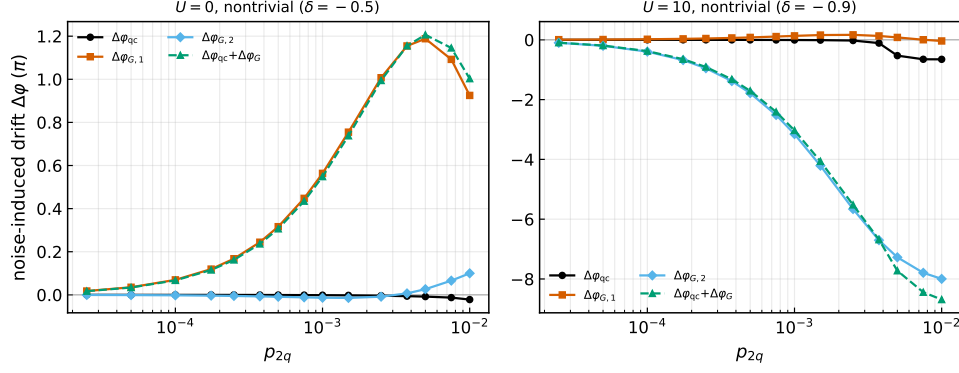


Figure 3: Noise-induced drift  $\Delta\varphi_x = \varphi_x(p_{2q}) - \varphi_x(0)$  of each leg of  $\varphi_B = \varphi_{qc} + \varphi_{G,1} + \varphi_{G,2}$  under increasing two-qubit error (exact-under-noise, no shots), plotted as the *change* from the noiseless value so the two regimes share a scale (the  $U = 10$  geometric leg carries an  $\approx 8\pi$  absolute winding that would otherwise swamp the plot). *Left* ( $U = 0$ ): the readout  $\varphi_{qc}$  and geometric leg  $\varphi_{G,2}$  barely move while the dynamical leg  $\varphi_{G,1} = -\int \langle \hat{H} \rangle dt$  drifts. *Right* ( $U = 10$ ): the readout again stays put, but now the *geometric* leg  $\varphi_{G,2}$  dominates the drift, with  $\varphi_{G,1}$  secondary. In both regimes the folded  $\varphi_B$  tracks the dominant integrated leg, never the readout.

**Shots vs. gate noise.** Figure 2 (left) separates the two axes. At negligible gate noise the phase-estimate spread falls as  $1/\sqrt{S}$  and the shots floor is low ( $S^* = 512$  shots for the  $U = 0$  nontrivial chain): finite sampling alone is not the bottleneck at the reference profile. Once incoherent noise is switched on, however, increasing  $S$  does not restore accuracy—the deviation saturates at a deterministic, gate-noise-induced bias that shots cannot average away. The method is therefore gate-noise-limited in the regime that matters.

## 6 Mechanism, failure, and cost analysis

**Does the noiseless cancellation survive noise?** Only partially, and the failure is diagnosable and *regime dependent*. We decompose the noise-induced drift into the three measured pieces (Fig. 3). One fact is common to both interaction strengths and both topological sectors: the single-ancilla readout  $\varphi_{qc}$  is the *most* robust piece, holding within a few percent of its noiseless value until the state is nearly maximally mixed ( $p_{2q} \gtrsim 2 \times 10^{-3}$ ). Its well-conditioned two-basis estimator is one (if deep) circuit whose error is bounded, not accumulated. The fragility instead lives in the two classically integrated global-phase legs, each a sum of  $O(10^2\text{--}10^4)$  device measurements taken node-by-node along the loop, where a small per-node bias accumulates rather than cancels. *Which* leg limits the threshold is set by the physics.

For the noninteracting chain the *dynamical* leg  $\varphi_{G,1} = -\int \langle \hat{H} \rangle dt$  dominates. Noiselessly the forward and time-reversed halves cancel this dynamical phase exactly (the Hamiltonian is time-reversal invariant); under noise, depolarizing and relaxation pull the measured  $\langle \hat{H} \rangle$  toward its maximally mixed value, and because the two halves accumulate different amounts of noise the cancellation breaks and the residual integrates into a drift (Fig. 3, left:  $\varphi_{G,1}$  swings while  $\varphi_{qc}, \varphi_{G,2}$  hold). For the correlated chain the *geometric* leg  $\varphi_{G,2} = \int \sum_\mu \langle \tilde{A}_\mu \rangle \dot{\theta}_\mu dt$  dominates instead. Its adaptive loop carries a large geometric winding,  $\varphi_{G,2} \approx +8.036\pi \approx 8\pi$ , measured on deep partial states; decoherence attenuates each connection overlap  $\langle \tilde{A}_\mu \rangle$  fractionally toward zero, and that fractional attenuation, multiplied by the large winding and summed over  $\sim 10^4$  nodes, produces a large absolute drift (Fig. 3, right:  $\varphi_{G,2}$  falls by several radians while  $\varphi_{G,1}$  moves only mildly and  $\varphi_{qc}$  holds). The geometric leg’s fragility thus scales with *both* circuit depth (more per-node attenuation) and winding magnitude—which is why the interacting thresholds sit roughly an order of magnitude below the noninteracting ones although the readout is no more fragile, and why the ordering is not a plain function of CNOT count.

The unifying statement: the limiting quantity is always an integral of node-by-node device measurements whose noise bias does not cancel over the loop, and never the readout—the regime only decides which integral is large. This is precisely the opposite of the noiseless picture, where the co-evolving legs cancel by construction. The coherent-error cancellation that gives the method its celebrated robustness to non-adiabaticity does not extend to the stochastic, node-accumulated bias that hardware noise injects.

**Which noise source dominates.** At the base profile’s fixed ratios the two-qubit depolarizing channel and  $T_1/T_2$  relaxation, both compounding with CNOT count and depth, drive the drift; readout error and single-qubit depolarizing are subdominant at the same  $\alpha$ . This is consistent with the interacting loops breaking roughly an order of magnitude before the noninteracting ones: it is the entangling-gate count and loop duration that set both the state’s decoherence and the per-node bias the  $\varphi_G$  integrals accumulate. Within a fixed  $U$  the nontrivial regime breaks first; across  $U$ , however, the threshold tracks the dominant integrated leg (energy bias at  $U = 0$ , geometric winding at  $U = 10$ ) rather than CNOT count alone, so the shallower  $U = 10$  nontrivial loop breaks marginally before the deeper  $U = 10$  trivial one.

**Computational cost.** The characterization is dominated by the  $\varphi_G$ -integrand measurements: each swept  $(\alpha, S, \text{seed})$  point requires, along the frozen trajectory, a noisy evaluation of  $\varphi_{qc}$  and of the  $\varphi_G$  integrand at  $O(10^2)$  integration nodes, each involving one energy measurement and  $N_\theta = O(10^2)$  connection overlaps, so the per-point cost scales as  $(\text{nodes}) \times N_\theta$  noisy circuit evaluations rather than a single readout. Because the incoherent-noise density matrix is deterministic given the rates, we compute it once per  $(\text{configuration}, \alpha)$  and reuse it across shot counts and seeds, with finite-shot noise added by real binomial draws; this factorization, single-threaded per worker and parallel across  $\alpha$ , makes the dense sweep tractable on a CPU node.

**Failure analysis.** Beyond threshold the estimate does not fail randomly: the mean phase migrates *smoothly* from the quantized value as the dominant integrated leg drifts, with the seed variance staying small until the state is nearly maximally mixed ( $p_{2q} \gtrsim 2 \times 10^{-3}$ ), where it finally blows up and the folded sign becomes undetermined. For the interacting chain this smooth migration is dramatic: because  $\varphi_{G,2}$  starts near  $8\pi$  and falls by tens of radians as noise grows, the folded  $\varphi_B$  wraps repeatedly around the circle, so the deviation-vs-noise curve is non-monotonic and can dip back below tolerance at strong noise (e.g. near  $p_{2q} \sim 10^{-3}$ , where the output infidelity already exceeds 0.7). Such a dip is a *coincidental* re-crossing of an essentially random folded phase, not a recovery of accuracy—which is exactly why the threshold rule was pre-registered to take the *first* upward crossing. The practical consequence is the same in both regimes: a naive run at current hardware noise returns a confident but wrong phase (a smooth intermediate value, or the wrong branch), not an obviously broken one—which is why the whole-method threshold, not the noiseless anchor, is the number that matters.

## 7 Related work

The estimator we characterize is the adaptive-variational Berry-phase method of [4], which combines the gate-based cyclic-adiabatic construction of [5] with AVQDS [12] and AVQITE [3]; its noiseless results are our anchor. Robust topological-invariant measurement on NISQ processors has been pursued by quench and randomized-measurement routes [7, 11] and via VQE-based topology detection [8]; our focus is orthogonal—not a new invariant estimator but a resilience characterization of this one. Our density-matrix channels are standard [2], and the finding that the *classically integrated* global-phase legs—not the readout—are the fragile part connects to error-bound analyses of variational time evolution [14]. Error mitigation [2] and metric-aware shot allocation [9] are the natural levers, and our mechanism analysis says where to aim them: at whichever integrated leg dominates for the regime—the energy integral  $\varphi_{G,1}$  for weak correlation, the geometric-connection integral  $\varphi_{G,2}$  for strong correlation.

## 8 Limitations and conclusions

Our study freezes the noiselessly grown ansatz and trajectory and injects noise into the readout and  $\varphi_G$ -integrand measurements; it does not feed sampling noise back into the McLachlan integration, so

it characterizes the resilience of the measured  $\varphi_B = \varphi_{qc} + \varphi_G$  at a fixed variational trajectory, not the noise-driven evolution of that trajectory (the optional noisy-construction study). The geometric connections are measured by the hardware-efficient partial-state primitive; a full-depth generalized Hadamard test would be more pessimistic. For  $U = 10$  the AVQDS ansatz is capped for tractability of the density-matrix sweep, leaving the McLachlan distance above  $L_{cut}^2$  in the fast-growth window near  $\lambda = 0$ ; the reproduced phase confirms a faithful gapped loop, but the interacting thresholds should be read with this caveat (fully uncapped runs are left to larger compute). We study one four-site chain, as the contract emphasizes characterization quality over system size.

The conclusion is sharp and, we believe, useful. The shallow circuits that make this adaptive-variational Berry-phase method attractive do not by themselves make its output noise-robust: the whole-method estimate—which the paper’s own construction requires, because  $\varphi_G$  is an integral of device-measured quantities—breaks at two-qubit error rates one to more than two orders of magnitude below current hardware ( $p_{2q}^* \approx 1.4 \times 10^{-4}$  for the noninteracting chain,  $\approx 2.6 \times 10^{-5}$  for the correlated one), and it does so through a specific, non-cancelling drift of a classically integrated global-phase leg rather than through the Hadamard readout. The noiseless robustness to non-adiabaticity, a coherent-error statement, does not carry over to stochastic noise. The corollary is constructive: because the fragility is concentrated in one identifiable, classically integrated quantity—and we say which one for each regime (the energy integral  $\varphi_{G,1}$  when correlations are weak, the large-winding geometric integral  $\varphi_{G,2}$  when they are strong)—mitigation targeted at that leg (zero-noise extrapolation of the offending integrand, or its cancellation by construction) is the most direct route to raising the threshold into the hardware-accessible range.

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